HTML Structure:

`! + Enter` → Basic HTML structure

<link> → Links external resources → **<link rel="stylesheet" href="style.css">**

Basic Website Structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
          content="width=device-width,
          initial-scale=1.0">
    <title>My Website</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <h1>Website Header</h1>
        <nav>
            <a href="#">Home</a>
            <a href="#">About</a>
            <a href="#">Contact</a>
        </nav>
    </header>
    <main>
        <section>
            <h2>Welcome</h2>
            <p>This is content...</p>
        </section>
    </main>
    <footer>
        <p>&copy; 2026 My Website</p>
    </footer>
</body>
</html>
```

HTML Text & Formatting:

<h1>-<h6> Headings.

<p> Paragraph.

**
** Line break.

<hr> Horizontal rule.

/<b> Bold.

/<i> Italic.

<u> Underline.

<small> Smaller text.

<sub>/<sup> Sub/Superscript.

<code> Code snippet.

<time> → **<time datetime="2026-08-28">**August 28, 2026**</time>**

HTML Containers

** ** Block-level container.

**** Inline container.

<section> Thematic grouping.

<nav> Navigation links.

<header> Intro/navigation.

<main> Main content.

<footer> Footer info.

HTML Lists

<ul> Unordered list.

<ol> Ordered list.

<li> List item. `value` for order-list →

```
<ol>
    <li value="1">Learn With Mahfuz</li>
</ol>
```

HTML Links & Media:

<a> Hyperlink → ****Visit****

<img> Image → **<img src="photo.jpg" alt="Profile" width="200" height="150">**

<table> Table wrapper.

<caption> Table caption.

<thead> Table header.

<tbody> Table body.

<tr> Table row.

<th> Header cell. `colspan`, `rowspan`, `headers`.

<td> Data cell. `colspan`, `rowspan`, `headers`.

```
<table border="1">
    <caption>Table Title</caption>
    <thead>
        <tr>
            <th>Header 1</th>
            <th>Header 2</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>Data 1</td>
            <td>Data 2</td>
        </tr>
    </tbody>
</table>
```

HTML Forms:

<form> Form wrapper.

<input> Input field: Type: text, password, email, number, url, checkbox, radio, date, time, datetime-local, color, file, range, submit, button

Input Attributes: `name`, `value`, `placeholder`, `required`, `disabled`, `min`, `max`

<button> Button. `type (submit/reset/button)`, `disabled`, `name`, `value`.

<select> Select/dropdown menu.

<option> Select option. `value`, `selected`, `disabled`.

<progress> Progress bar. `value`, `max`.

```
<form>
    <label>Email:</label>
    <input type="email" placeholder="Enter your email">
    <br>
    <label>Password:</label>
    <input type="password">
</form>

<progress value="75" max="100"></progress>

<select>
    <option value="cse">CSE</option>
    <option value="eee">EEE</option>
</select>
```

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CSS Quick Reset:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
```

CSS Selectors:

p { } → Element/Tag selector. Selects all <p> elements.
.class { } → Class selector. Selects elements with a specific `class`.
#id { } → ID selector. Selects the element with a specific `id`.
*** { }** → Universal selector. Selects all elements.
div p { } → Descendant selector. Selects all <p> elements inside .
div > p { } → Child selector. Selects <p> elements that are direct children of .
h1, h2, p { } → Grouping selector. Selects multiple elements.

CSS Box Model:

width — Element width.
height — Element height.
min-width — Minimum width limit.
max-width — Maximum width limit.
min-height — Minimum height limit.
max-height — Maximum height limit.
padding — Space inside the border.
padding-top/right/bottom/left — Individual padding.
border — Line around the element.
border-width — Border thickness.
border-style — Border style: solid, dashed, dotted, etc.
border-color — Border color.
margin — Space outside the border.
margin-top/right/bottom/left — Individual margin.
box-sizing: border-box — Includes padding and border within width/height.
border-radius — Rounded corners.

CSS Text & Font:

font-family — Font type. Example: Arial, sans-serif.
font-size — Text size. Example: px, em, rem, %.
font-weight — Text thickness. Example: normal, bold, 100–900.
font-style — Text style. Example: normal, italic.
text-align — Horizontal text alignment: left, center, right, justify.
text-decoration — Text decoration: none, underline, line-through.

CSS Colors & Opacity:

color — Text color. Example: red, #ff5733.
rgb() — Defines color using RGB values. Example: rgb(255, 87, 51).
opacity — Element transparency. Range: 0 to 1.

CSS Backgrounds:

background-color — Background fill color.
background-image — Sets a background image. Example: url("image.jpg").
background: linear-gradient(to right, blue, yellow) — Linear gradient background.
background: radial-gradient(ellipse, gold, blue) — Radial gradient background.
background-repeat — Controls image repetition: repeat, no-repeat, repeat-x, repeat-y.
background-size — Controls image size: cover, contain, 100px, etc.

CSS Display & Positioning:

display — Controls how an element is displayed: block, inline, inline-block, none.
position — Sets positioning method: static, relative, absolute, fixed, sticky.
top/right/bottom/left — Position offsets for positioned elements.
z-index — Controls stack order.

CSS Flexbox:

display: flex — Enables Flexbox.
flex-direction — Sets the main axis: row, row-reverse, column, column-reverse.
justify-content — Aligns items along the main axis: flex-start, center, flex-end, space-between, space-around, space-evenly.
align-items — Aligns items along the cross axis: stretch, center, flex-start, flex-end, baseline.
align-content — Aligns multiple flex lines: flex-start, center, flex-end, space-between, space-around, space-evenly, stretch.
flex-wrap — Controls wrapping: nowrap, wrap, wrap-reverse.
gap — Sets space between items. Example: 10px or 10px 20px.

CSS Grid:

display: grid — Enables Grid layout.
grid-template-columns — Defines grid columns. Example: repeat(3, 1fr).
grid-template-rows — Defines grid rows. Example: auto 100px.
gap — Sets space between rows and columns. Example: 20px or 10px 20px.
row-gap — Sets space between rows.
column-gap — Sets space between columns.
justify-items — Aligns grid items horizontally within their grid areas.
align-items — Aligns grid items vertically within their grid areas.

CSS Shadow & Effects:

box-shadow — Adds a shadow around an element. Example: 5px 5px 10px rgba(0,0,0,0.3).
box-shadow: inset ... — Creates an inner shadow.
text-shadow — Adds a shadow to text. Example: 2px 2px 4px black.

CSS Overflow:

overflow — Controls content that exceeds an element's box: visible, hidden, scroll, auto.
overflow-x — Controls horizontal overflow.
overflow-y — Controls vertical overflow.
overflow-wrap — Controls whether long words/strings can break onto a new line: normal, break-word, anywhere.

CSS Relative Units:

vh — Viewport height. 1vh = 1% of viewport height.
vw — Viewport width. 1vw = 1% of viewport width.
% — Percentage relative to the parent/container or relevant property.

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